

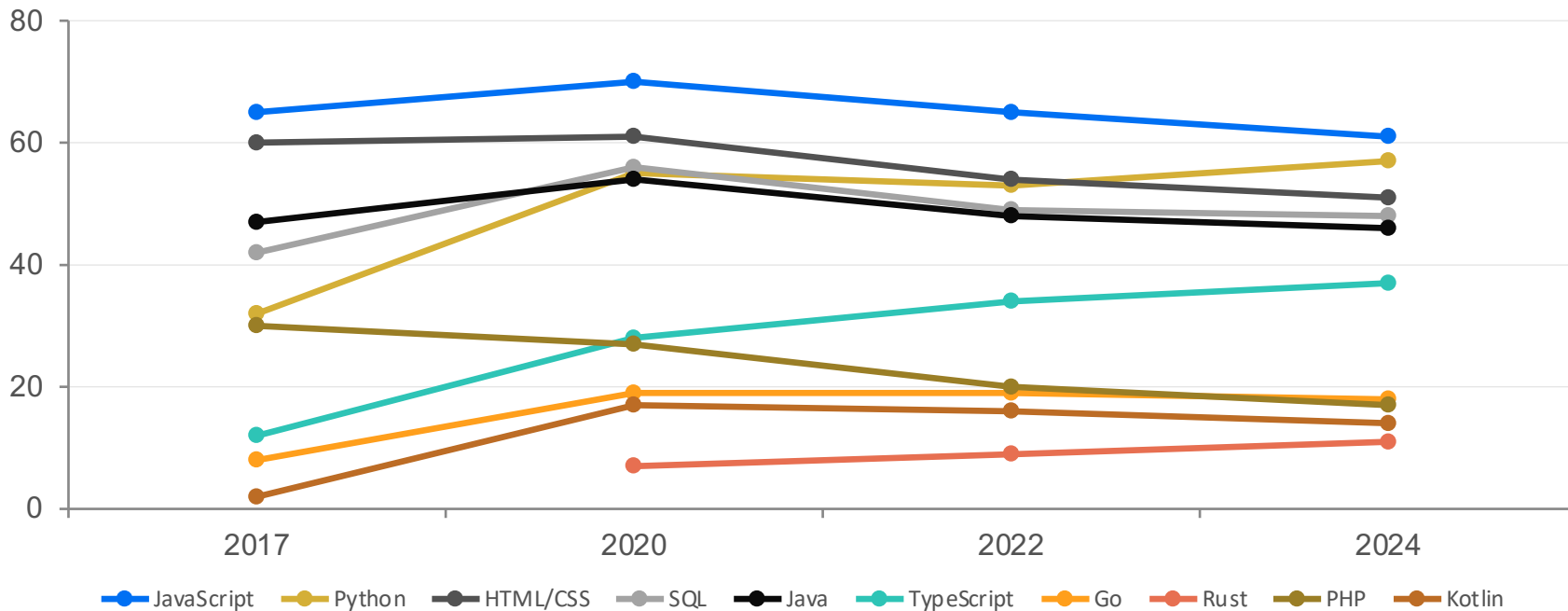
State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

[JetBrains State of Developer Ecosystem Report 2024](#)

Python Closes the Gap: 8 Years of Language Usage Shifts

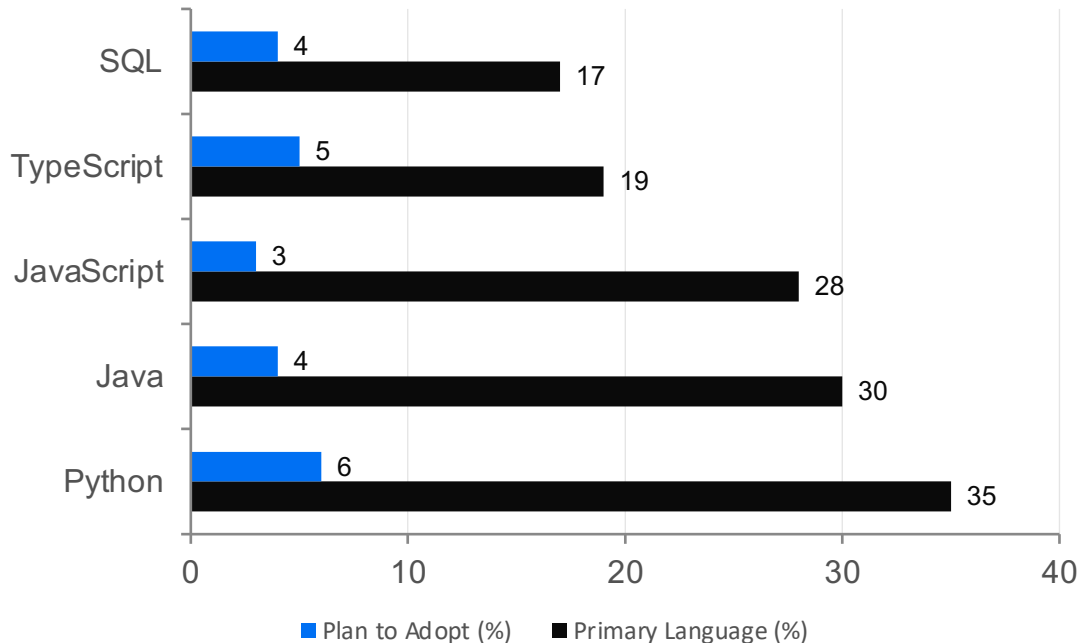
Usage share of top programming languages, 2017-2024 (% of respondents)



Source: JetBrains State of Developer Ecosystem Report 2024 (n = 23,262)

Python Is Now the #1 Primary Language at 35%

Primary language share vs adoption intent — developers rarely switch stacks



Key Insight

Ecosystem lock-in is real.

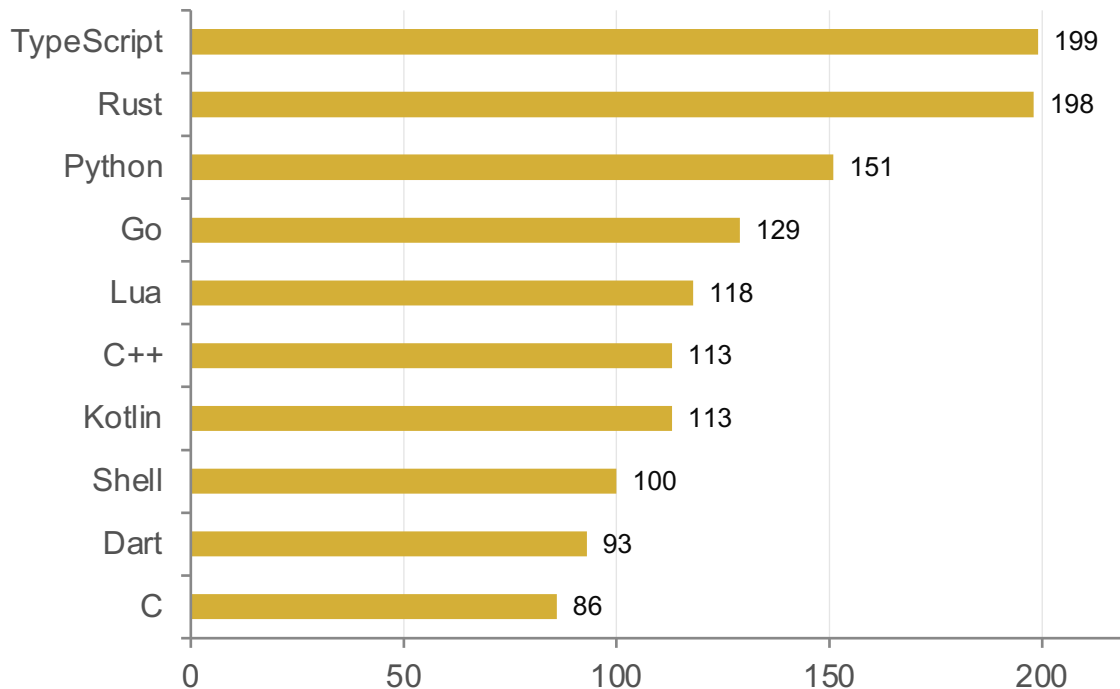
Adoption intent for new primary languages remains below 6% across the board. Developers stick with what they know, making hiring and retention critical.

Python overtakes JavaScript as the most common primary language (35%), reflecting AI/ML and data-science dominance.

Source: JetBrains State of Developer Ecosystem Report 2024

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Language satisfaction scores — type-safe, modern-tooling languages dominate the top spots



Source: JetBrains State of Developer Ecosystem Report 2024

TypeScript — 199

Strong type safety + modern tooling + expressive type systems drive enthusiasm.

Rust — 198

Memory safety without garbage collection; compiler guarantees reduce bugs and boost confidence.

Python — 151

Third place despite massive usage; satisfaction is high but the gap with TS/Rust reveals DX expectations.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by domain, not general preference — match hiring to the problem space

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Science / Analytics	Python	28%
User Interface Development	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

Source: JetBrains State of Developer Ecosystem Report 2024

Strategic Implication

Hiring should map to domains, not just language popularity.

- AI/ML teams → Python talent pool
- Frontend / full-stack → TypeScript
- Enterprise APIs → Java
- Systems / infra → C++ or Rust

What This Means for Your Technology Strategy in 2025

01

Invest in Python for AI/ML teams

Python is the undisputed leader in AI (31%), data science (28%), and data processing. Build your AI strategy around this talent pool.

02

TypeScript is the future of frontend

With 62% share in UI development and the highest satisfaction score (199), TypeScript should be your default for web and full-stack.

03

Rust and Go are infrastructure growth plays

Go (18% usage, plateaued) and Rust (11%, climbing) are the top adoption targets for microservices, cloud-native, and systems programming.

04

PHP and Ruby hiring pools are shrinking

PHP dropped from 30% to 17%; Ruby fell from 10% to 4%. Plan migration or reduced reliance on legacy web stacks.

05

Satisfaction correlates with retention

Source: JetBrains State of Developer Ecosystem Report 2024 | n = 23,262 developers worldwide

TypeScript and Rust lead satisfaction thanks to type safety and modern tooling. Factor developer experience into retention strategy.